Machine Learning, 50-day challenge

Day 1 pre-reading guide

Is this an ML problem? Learn these six things first, then test yourself before you attempt the challenge.

Total preparation time: about 3 to 4 hours. Do the resources in the order below.

The reading path

#	Resource	What to take from it	Link and time
1	**Google: ML Problem Framing**	How to check that ML is a good approach, then define the outcome, model output and success metrics. Includes framing exercises.	developers.google.com/machine-learning/problem-framing *Read first, 45–60 min*
2	**Rules of Machine Learning (Zinkevich)**	Read only the 'Before Machine Learning' section. Rule #1: a product can launch without ML, and a heuristic often gets about half the benefit.	developers.google.com/machine-learning/guides/rules-of-ml *15 min*
3	**The First Rule of ML: Start without ML (Eugene Yan)**	Why ML needs data, pipelines and good labels, with advice from practitioners who try a non-ML approach first.	eugeneyan.com/writing/first-rule-of-ml *10 min*
4	**Google ML Crash Course: classification metrics**	Why accuracy misleads on imbalanced data. Interactive exercises on precision, recall, ROC and AUC.	developers.google.com/machine-learning/crash-course/classification/precision-and-recall *45 min*
5	**scikit-learn: Choosing the right estimator**	Flowchart of problem types: classification, regression, clustering, dimensionality reduction.	scikit-learn.org/stable/machine_learning_map.html *15 min*
6	**Forecasting: Principles and Practice (Hyndman and Athanasopoulos)**	Free online book. Read the opening chapters and the simple benchmark methods, which matter for Problem 3.	otexts.com/fpp3 *60 min*

Self-test before you start

Answer these without looking anything up. If you get stuck on one, go back to the resource in brackets.

- ☐ Why can a model with 99.9% accuracy be useless for fraud detection? (Resource 4)
- ☐ Give one problem where a rule is better than ML, and explain why. (Resources 2 and 3)
- ☐ What is the difference between classification and forecasting, and why can't you shuffle time-series data? (Resources 5 and 6)
- ☐ What does clustering need from your data that classification needs, and what does it not need? (Resource 5)

- ☐ What is a baseline, and why must a model beat it? (Resources 1, 2 and 6)
- ☐ Name the two steps of framing a problem for ML. (Resource 1)

Ready? If you can answer all six in your own words, you are ready for the Day 1 challenge. If not, reread the matching resource and try again.

Also worth knowing

- The checklist on the Day 1 handout has seven questions. Keep it beside you while you read, and notice which resource answers which question.
- Problem 5 on the handout is about scheduling with rules. For background, search for the Google OR-Tools documentation on employee scheduling. This link has not been verified, so check that the page opens.
- Links were checked when this guide was prepared. If one does not open, search for the resource title.